1. About half of households do not hold any equity

   This is called the *stock non-participation puzzle*, and I discuss it further in chapter 3.
2. Older households hold bigger equity allocations than do young, working households.

   Another way to put this is that household equity shares are constant or even *increase* with age. Even at retirement, the equity fraction is very high. In retirement, households hang onto their equity holdings for as long as possible instead of moving as soon as they can to bonds and other low-risk assets.
3. The rich have larger equity weights than the poor.

There are also three stylized facts of households' consumption over their life cycles:[27]

1. Consumption is very smooth.

   Life-cycle models predict that when an individual's permanent income increases suddenly, the individual should consume more—after all, you've just enjoyed a windfall. But spending does not respond very fast to changes in income. This is called the *excess smoothness puzzle* in the literature.
2. At the same time, consumption is *excessively sensitive*.

   Although consumption doesn't move when projected income changes, consumption does react—and it reacts too much—to past changes in income.
3. Consumption is hump-shaped over the life cycle.

   The peak consumption years are in middle age. At retirement, consumption declines in line with lower labor income. Figure 5.6 shows consumption expenditures in different age brackets from 1984 to 2011 from the Consumption Expenditure Survey conducted by the Bureau of Labor Statistics. I have normalized the forty-five to fifty-four age bracket, where consumption peaks, at 100%. Note that the more recent retirees have started to consume more compared to those who retired in the 1980s.

It is an exciting time to be working in the life-cycle area. The newest models can explain these patterns by incorporating many realistic features: they take into account the inability to borrow (especially by the young), use utility functions that capture salient consumer behaviors, and model complex dynamics of labor income and asset returns.[28] These models can match the stylized facts of what people do

[27] This is a much larger literature than the life-cycle portfolio choice literature. The canonical references are Deaton (1991), Carroll (2001), and Gourinchas and Parker (2002). A literature summary is Attanasio and Weber (2010). Most of these papers, however, do not have portfolio choice decisions involving risky assets.

[28] For life-cycle models with financial constraints, see Rampini and Viswanathan (2013). Laibson, Repetto, and Tobacman (2012) and Pagel (2012) develop behavioral life-cycle models with hyperbolic